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\\MARTINOS DEVELOPER

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2mm_run5																	
mesovein_OR																	

\\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\AAhead_scout

TA: 14 sec Coil Selection: Auto Voxel Size: 1.6×1.6×1.6 mm³ Acc:: 3 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
TE	1.53 ms
Averages	1
Concatenations	1
AutoAlign	Head

Contrast - Common

TR	3.25 ms
TE	1.53 ms
Flip Angle	16 deg
Fat-Water Contrast	Standard
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Time to Center	6.3 s

Resolution - Common

FOV Read	260 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	1.6 mm
Base Resolution	160
Phase Resolution	100 %
Slice Resolution	69 %
Trajectory	Cartesian

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference Scans	Integrated
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	1
Phase Partial Fourier	6/8
Slice Partial Fourier	6/8
Asymmetric Echo	Weak

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	3D
Normalize	B1 Filter
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	128
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	260 mm
FOV Phase	100.0 %
Slice Thickness	1.6 mm
TR	3.25 ms
Multi-Slice Mode	Sequential
Series	Ascending
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.0 A30.0 H0.0 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	Head
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm

Geometry - AutoAlign

H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Subtraction

Subtract	Off
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Inline - Subtraction

Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Composing**Inline - MapIt**

MapIt	None
Flip Angle	16 deg
Measurements	1
Contrasts	1
TE	1.53 ms
TR	3.25 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Bandwidth	540 Hz/Px
Asymmetric Echo	Weak

Sequence - Part 2

Introduction	On
RF Spoiling	On
Breast Application	Off
Phase Enc. Order	Automatic

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\wip_ShimToolbox_B0map_3mm

TA: 44 sec Coil Selection: Auto Voxel Size: 3.0x3.0x3.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.8 A22.7 F34.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
Concatenations	1
AutoAlign	---

Contrast - Common

TR	15.0 ms
TE 1	2.20 ms
TE 2	5.40 ms
MTC	Off
Magn. Preparation	None
Flip Angle	8 deg
Fat-Water Contrast	Water Excitation
Dark Blood	Off
Contrasts	2
SWI	Off
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
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Contrast - Dynamic

Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
Base Resolution	72
Phase Resolution	100 %
Slice Resolution	100 %

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off
Slice Partial Fourier	Off
Asymmetric Echo	Off
Elliptical Scanning	On

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Distance Factor	20 %
Position	L0.8 A22.7 F34.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slices per Slab	52
Phase Oversampling	0 %
Slice Oversampling	0.0 %
FOV Read	216 mm
FOV Phase	100.0 %
Slice Thickness	3.00 mm
TR	15.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Concatenations	1

Geometry - AutoAlign

Slab Group	1
Position	L0.8 A22.7 F34.8 mm
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---

Geometry - AutoAlign

Initial Position	L0.8 A22.7 F34.8
L	0.8 mm
A	22.7 mm
F	34.8 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	L0.8 A22.7 F34.8 mm
Orientation	Sagittal
Rotation	0.00 deg
A >> P	216 mm
F >> H	216 mm
R >> L	156 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Non-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	297.136279 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000
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Physio - Signal

1st Signal/Mode	None
TR	15.0 ms
Segments	1
Concatenations	1

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Water Excitation
Magn. Preparation	None
Dark Blood	Off
FOV Read	216 mm
FOV Phase	100.0 %
Phase Resolution	100 %

Physio - PACE

Resp. Control	Off
Concatenations	1

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1
StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing**Inline - MapIt**

MapIt	None
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Inline - MapIt

Flip Angle	8 deg
Measurements	1
Contrasts	2
TE 1	2.20 ms
TE 2	5.40 ms
TR	15.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	WIP_GRE
Dimension	3D
Excitation	Non-sel.
RF Pulse Type	Normal
Readout Mode	Monopolar
Gradient Mode	Whisper
Flow Compensation 1	None
Flow Compensation 2	None
Bandwidth 1	500 Hz/Px
Bandwidth 2	500 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	Off
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Nuclei

TX/RX Nucleus	1H
TX/RX Delta Frequency	0 Hz
TX Nucleus	None
TX Delta Frequency	0 Hz

Sequence - Special

Mode	B0 Mapping
Masking	Brain Extraction
Phase Unwrapping	Spatial

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\1mm_run1

TA: 4:43 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Multi-echo spacing	42.40 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	180
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	128
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P57.7 H1.5
L	0.1 mm

Geometry - AutoAlign

P	57.7 mm
H	1.5 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P59.6 H4.3 mm
! Orientation	C > T-14.8 > S-0.4
! Rotation	-89.92 deg
! F >> H	153 mm
! R >> L	133 mm
! A >> P	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.04 ms
Turbo Factor	60
Segmentation	2
EPI Factor	40

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	800 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\mesovein_OR

TA: 6:27 min Coil Selection: Manual Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R11.5 P11.4 H12.3 mm
Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	2.9 %
Slice Oversampling	0.0 %
FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
TR	60.0 ms
Vol. TR	380.16 s
TE 1	10.60 ms
TE 2	27.20 ms
TE 3	43.80 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	60.0 ms
Vol. TR	380.16 s
TE 1	10.60 ms
TE 2	27.20 ms
TE 3	43.80 ms
Multi-echo spacing	16.60 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Magn. Prep. Shots	1
Contrasts	3
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
Base Resolution	466
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R11.5 P11.4 H12.3 mm
Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	2.9 %
Slice Oversampling	0.0 %
FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
TR	60.0 ms
Vol. TR	380.16 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R11.5 P11.4 H12.3 mm

Geometry - AutoAlign

Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R11.5 P11.4 H12.3
R	11.5 mm
P	11.4 mm
H	12.3 mm
Initial Orientation	T > C
T > C	-28.80
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	R4.3 P9.7 H6.1 mm
I Orientation	T > C-31.0
I Rotation	90.00 deg
I R >> L	127 mm
I A >> P	154 mm
I F >> H	98 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	466 Hz/Px
Echo Spacing	2.83 ms
Turbo Factor	6336
Segmentation	36
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	20
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.10
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10/10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\2mm_run1

TA: 4:28 min Coil Selection: Manual Voxel Size: 2.0x2.0x2.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Multi-echo spacing	37.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
Base Resolution	90
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P1.1 F19.6
L	0.1 mm

Geometry - AutoAlign

P	1.1 mm
F	19.6 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L5.6 P18.0 H4.9 mm
! Orientation	C > T28.0 > S-0.4
! Rotation	-89.60 deg
! F >> H	121 mm
! R >> L	133 mm
! A >> P	160 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1984 Hz/Px
Echo Spacing	0.59 ms
Turbo Factor	57
Segmentation	1
EPI Factor	62

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	440 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\mesovein_RO

TA: 6:27 min Coil Selection: Manual Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R11.5 P11.4 H12.3 mm
Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	2.9 %
Slice Oversampling	0.0 %
FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
TR	60.0 ms
Vol. TR	380.16 s
TE 1	10.60 ms
TE 2	27.20 ms
TE 3	43.80 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	60.0 ms
Vol. TR	380.16 s
TE 1	10.60 ms
TE 2	27.20 ms
TE 3	43.80 ms
Multi-echo spacing	16.60 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Magn. Prep. Shots	1
Contrasts	3
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
Base Resolution	466
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R11.5 P11.4 H12.3 mm
Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	2.9 %
Slice Oversampling	0.0 %
FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
TR	60.0 ms
Vol. TR	380.16 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R11.5 P11.4 H12.3 mm

Geometry - AutoAlign

Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R11.5 P11.4 H12.3
R	11.5 mm
P	11.4 mm
H	12.3 mm
Initial Orientation	T > C
T > C	-28.80
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	R4.3 P9.7 H6.1 mm
I Orientation	T > C-31.0
I Rotation	90.00 deg
! R >> L	127 mm
! A >> P	154 mm
! F >> H	98 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	466 Hz/Px
Echo Spacing	2.83 ms
Turbo Factor	6336
Segmentation	36
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	20
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.10
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10/10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Regular RO

Sequence - Assistant

SAR Assistant	Off
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\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\1mm_run2

TA: 4:43 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Multi-echo spacing	42.40 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	180
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	128
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P57.7 H1.5
L	0.1 mm

Geometry - AutoAlign

P	57.7 mm
H	1.5 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P59.6 H4.3 mm
! Orientation	C > T-14.8 > S-0.4
! Rotation	-89.92 deg
! F >> H	153 mm
! R >> L	133 mm
! A >> P	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.04 ms
Turbo Factor	60
Segmentation	2
EPI Factor	40

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	800 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\2mm_run2

TA: 4:28 min Coil Selection: Manual Voxel Size: 2.0x2.0x2.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Multi-echo spacing	37.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
Base Resolution	90
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P1.1 F19.6
L	0.1 mm

Geometry - AutoAlign

P	1.1 mm
F	19.6 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L5.6 P18.0 H4.9 mm
! Orientation	C > T28.0 > S-0.4
! Rotation	-89.60 deg
! F >> H	121 mm
! R >> L	133 mm
! A >> P	160 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1984 Hz/Px
Echo Spacing	0.59 ms
Turbo Factor	57
Segmentation	1
EPI Factor	62

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	440 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\1mm_run3

TA: 4:43 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Multi-echo spacing	42.40 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	180
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	128
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P57.7 H1.5
L	0.1 mm

Geometry - AutoAlign

P	57.7 mm
H	1.5 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P59.6 H4.3 mm
! Orientation	C > T-14.8 > S-0.4
! Rotation	-89.92 deg
! F >> H	153 mm
! R >> L	133 mm
! A >> P	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.04 ms
Turbo Factor	60
Segmentation	2
EPI Factor	40

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	800 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\2mm_run3

TA: 4:28 min Coil Selection: Manual Voxel Size: 2.0x2.0x2.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Multi-echo spacing	37.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
Base Resolution	90
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P1.1 F19.6
L	0.1 mm

Geometry - AutoAlign

P	1.1 mm
F	19.6 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L5.6 P18.0 H4.9 mm
! Orientation	C > T28.0 > S-0.4
! Rotation	-89.60 deg
! F >> H	121 mm
! R >> L	133 mm
! A >> P	160 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1984 Hz/Px
Echo Spacing	0.59 ms
Turbo Factor	57
Segmentation	1
EPI Factor	62

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	440 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\1mm_run4

TA: 4:43 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Multi-echo spacing	42.40 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	180
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	128
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P57.7 H1.5
L	0.1 mm

Geometry - AutoAlign

P	57.7 mm
H	1.5 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P59.6 H4.3 mm
! Orientation	C > T-14.8 > S-0.4
! Rotation	-89.92 deg
! F >> H	153 mm
! R >> L	133 mm
! A >> P	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.04 ms
Turbo Factor	60
Segmentation	2
EPI Factor	40

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	800 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 /10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\2mm_run4

TA: 4:28 min Coil Selection: Manual Voxel Size: 2.0x2.0x2.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Multi-echo spacing	37.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
Base Resolution	90
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P1.1 F19.6
L	0.1 mm

Geometry - AutoAlign

P	1.1 mm
F	19.6 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L5.6 P18.0 H4.9 mm
! Orientation	C > T28.0 > S-0.4
! Rotation	-89.60 deg
! F >> H	121 mm
! R >> L	133 mm
! A >> P	160 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1984 Hz/Px
Echo Spacing	0.59 ms
Turbo Factor	57
Segmentation	1
EPI Factor	62

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	440 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\1mm_run5

TA: 4:43 min Coil Selection: Manual Voxel Size: 1.0x1.0x1.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	50.0 ms
Vol. TR	3000 ms
TE 1	20.00 ms
Multi-echo spacing	42.40 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
Base Resolution	180
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	128
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	7/8
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	60
Phase Oversampling	0.0 %
Slice Oversampling	0.0 %
FOV Read	180 mm
FOV Phase	100.0 %
Slice Thickness	1.00 mm
TR	50.0 ms
Vol. TR	3000 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P57.7 H1.5 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P57.7 H1.5
L	0.1 mm

Geometry - AutoAlign

P	57.7 mm
H	1.5 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L0.0 P59.6 H4.3 mm
! Orientation	C > T-14.8 > S-0.4
! Rotation	-89.92 deg
! F >> H	153 mm
! R >> L	133 mm
! A >> P	60 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1112 Hz/Px
Echo Spacing	1.04 ms
Turbo Factor	60
Segmentation	2
EPI Factor	40

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	800 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\2mm_run5

TA: 4:28 min Coil Selection: Manual Voxel Size: 2.0x2.0x2.0 mm³ Acc:: 4 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	52.6 ms
Vol. TR	2998.2 ms
TE 1	20.00 ms
Multi-echo spacing	37.20 ms
MTC	Off
Flip Angle	18 deg
Fat-Water Contrast	Fast Fat Saturation
Magn. Prep. Shots	1
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	88
Reordering	Linear

Resolution - Common

FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
Base Resolution	90
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	2
Reference Lines PE	48
Acceleration Factor 3D	2
Reference Lines 3D	40
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
Slices per Slab	96
Phase Oversampling	0.0 %
Slice Oversampling	18.8 %
FOV Read	180 mm
FOV Phase	137.8 %
Slice Thickness	2.00 mm
TR	52.6 ms
Vol. TR	2998.2 ms
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	L0.1 P1.1 F19.6 mm
Orientation	C > T-13.9
Phase Encoding Dir.	H >> F
AutoAlign	---
Initial Position	L0.1 P1.1 F19.6
L	0.1 mm

Geometry - AutoAlign

P	1.1 mm
F	19.6 mm
Initial Orientation	C > T
C > T	-13.90
> S	0.00
Initial Rotation	-90.00 deg

Geometry - Saturation

Saturation Mode	Quick
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

! Position	L5.6 P18.0 H4.9 mm
! Orientation	C > T28.0 > S-0.4
! Rotation	-89.60 deg
! F >> H	121 mm
! R >> L	133 mm
! A >> P	160 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
! Ref. Amplitude 1H	250.000 V
Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	1984 Hz/Px
Echo Spacing	0.59 ms
Turbo Factor	57
Segmentation	1
EPI Factor	62

Sequence - Part 2

Introduction	Off
RF Spoiling	On

Sequence - Special

RF duration	440 us
RF time x BW	10
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	On
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.20
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	100 / 10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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\\MARTINOS DEVELOPER\HUBER\Shahin_Kaisu\Shahin_20260708_1mm_2mm\mesovein_OR

TA: 6:27 min Coil Selection: Manual Voxel Size: 0.4x0.4x0.4 mm³ Acc:: 6 Rel. SNR: 1.00**Properties**

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slab Group	1
Slabs	1
Position	R11.5 P11.4 H12.3 mm
Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	2.9 %
Slice Oversampling	0.0 %
FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
TR	60.0 ms
Vol. TR	380.16 s
TE 1	10.60 ms
TE 2	27.20 ms
TE 3	43.80 ms
Averages	1
Multi-echo Shots	1
AutoAlign	---
Coil Elements	AC

Contrast - Common

TR	60.0 ms
Vol. TR	380.16 s
TE 1	10.60 ms
TE 2	27.20 ms
TE 3	43.80 ms
Multi-echo spacing	16.60 ms
MTC	Off
Flip Angle	15 deg
Fat-Water Contrast	Standard
Magn. Prep. Shots	1
Contrasts	3
Reconstruction	Magn./Phase

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Reordering	Linear

Resolution - Common

FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
Base Resolution	466
Phase Resolution	100 %
Slice Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	CAIPIRINHA
Reference Scans	GRE/Separate
Acceleration Factor PE	3
Reference Lines PE	24
Acceleration Factor 3D	2
Reference Lines 3D	24
Reordering Shift 3D	1
Phase Partial Fourier	Off
Slice Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Normalize	Off
Image Filter	Off

Geometry - Common

Slab Group	1
Slabs	1
Position	R11.5 P11.4 H12.3 mm
Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
Slices per Slab	352
Phase Oversampling	2.9 %
Slice Oversampling	0.0 %
FOV Read	165 mm
FOV Phase	112.7 %
Slice Thickness	0.35 mm
TR	60.0 ms
Vol. TR	380.16 s
Multi-echo Shots	1

Geometry - AutoAlign

Slab Group	1
Position	R11.5 P11.4 H12.3 mm

Geometry - AutoAlign

Orientation	T > C-28.8
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	R11.5 P11.4 H12.3
R	11.5 mm
P	11.4 mm
H	12.3 mm
Initial Orientation	T > C
T > C	-28.80
> S	0.00
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
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Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H

System - Miscellaneous

Coil Selection	Manual
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Brain
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

I Position	R4.3 P9.7 H6.1 mm
I Orientation	T > C-31.0
I Rotation	90.00 deg
I R >> L	127 mm
I A >> P	154 mm
I F >> H	98 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slab-sel.

System - Tx/Rx

Frequency 1H	297.136279 MHz
? Ref. Amplitude 1H	0.000 V

System - Tx/Rx

Reset	Off
Correction Factor	1.00
Image Scaling	1.000

Sequence - Part 1

Sequence Name	vaso a2d6453
Dimension	3D
Excitation	Slab-sel.
RF Pulse Type	Normal
Gradient Mode	Normal
Reordering	Linear
Bandwidth	466 Hz/Px
Echo Spacing	2.83 ms
Turbo Factor	6336
Segmentation	36
EPI Factor	5

Sequence - Part 2

Introduction	On
RF Spoiling	On

Sequence - Special

RF duration	1000 us
RF time x BW	20
PAT ref. FA	3 deg
Fat sat. FA	110 deg
T1 (Ernst FA)	1200 ms
Invert PE	Off
Min. TE w/ PF	On
Ramp Sampling	On
Trigger per shot	Off
Noise image	Off
Relax spoilers	Off
Round up Vol. TR	Off
MT flip angle	500 deg
MT off-res.	2000 Hz
MT RF duration	10240 us
Custom Water Exc.	-none-
Phase Correction	per Series
Saturation RF	per Shot
EPI rise time factor	1.10
G. spoil dephasing[1]	0.0 pi
G. spoil dephasing[2]	4.0 pi
G. spoil dephasing[3]	2.0 pi
Modify Ice Config	On
G-factor map	Off
GRAPPA Regularization	10/10^6
Slab Scale	-10 %
RF spoil scheme	Conventional
Read polarity	Inverted RO

Sequence - Assistant

SAR Assistant	Off
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